

Two Qubit circuit

tensor product:

$$A = \begin{bmatrix} a \\ b \end{bmatrix} \quad B = \begin{bmatrix} c \\ d \end{bmatrix}$$

$$A \otimes B = \begin{bmatrix} aB \\ bB \end{bmatrix} = \begin{bmatrix} ac \\ ad \\ bc \\ bd \end{bmatrix}$$

$$\begin{array}{l} |0\rangle \text{ —————} \\ |0\rangle \text{ —————} \end{array}$$

$$|0\rangle \otimes |0\rangle = |00\rangle$$

$$|00\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$|01\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$|11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad \text{Binary}$$

$$\begin{array}{l} |\psi\rangle \text{ --- } \\ |\delta\rangle \text{ --- } \end{array}$$

$$\begin{aligned} |\psi\rangle &= a|0\rangle + b|1\rangle \\ |\delta\rangle &= c|0\rangle + d|1\rangle \end{aligned}$$

$$|\psi\rangle|\delta\rangle = (a|0\rangle + b|1\rangle)(c|0\rangle + d|1\rangle)$$

$$= ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

$$p(d) = \|ad\|^2$$

Applying gates:

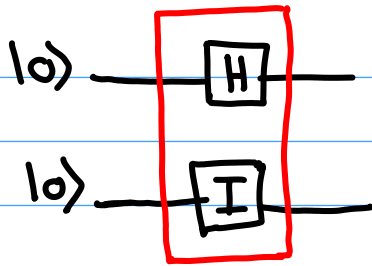
$$\begin{array}{l} |0\rangle \text{ --- } \boxed{H} \text{ --- } \\ |0\rangle \text{ --- } \end{array}$$

$$|0\rangle \otimes |0\rangle$$

$$= H|0\rangle \otimes |0\rangle$$

$$= |+\rangle |0\rangle$$

$$= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |0\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$$



$$U = H \otimes I$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$(H \otimes I) |00\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \frac{|00\rangle + |10\rangle}{\sqrt{2}}$$

$$H|0\rangle \otimes |0\rangle = |+\rangle|0\rangle$$



$$H \otimes H |00\rangle = |+\rangle|+\rangle$$

$$= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$= \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

Measurement:



$|0\rangle$

Apply M

$$p(0)=1$$

output = $|0\rangle$

$$|\psi\rangle = a|0\rangle + b|1\rangle \quad \text{Apply } M$$

$$p(0) = |a|^2$$

$$p(1) = |b|^2$$

$$\text{shots} = N$$

$$|0\rangle = |a|^2 N$$

$$|1\rangle = |b|^2 N$$



$|00\rangle$

$|10\rangle$

$$\frac{1}{\sqrt{2}} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

X on q_0

H on q_1

Apply M on q_0

$$p(0) \text{ on } q_0 = 0$$

$$p(1) \text{ on } q_0 = 1$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

Apply M on q_1

$$p(0) \text{ on } q_1 = \frac{1}{2}$$

$$p(1) \text{ on } q_1 = \frac{1}{2}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} |10\rangle$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} |11\rangle$$

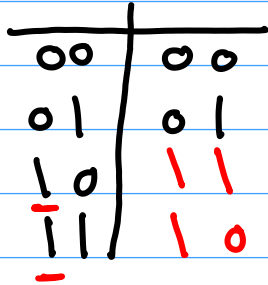
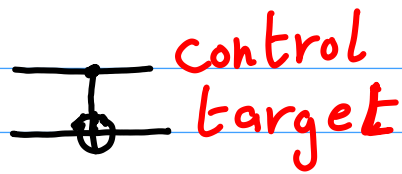
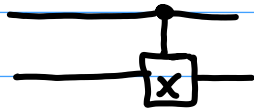
$$\sqrt{\left|\frac{1}{\sqrt{2}}\right|^2} = \frac{1}{\sqrt{2}}$$

$$|\psi\rangle = |10\rangle$$

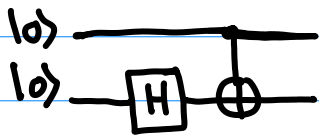
$$|\psi\rangle = |11\rangle$$

2-qubit gates:

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}_{4 \times 4}$$



cannot be $A \otimes B$



$|00\rangle$
 $|0\rangle|+\rangle$
 $|0\rangle|+\rangle$

H on q_1
 CNOT q_0, q_1

$|10\rangle$
 $|1\rangle|+\rangle$

H on q_1
 CNOT q_0, q_1

$$\frac{1}{\sqrt{2}} |1\rangle (|0\rangle + |1\rangle)$$

$$\frac{1}{\sqrt{2}} |10\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

$$\frac{1}{\sqrt{2}} |11\rangle + \frac{1}{\sqrt{2}} |10\rangle$$

$$CNOT = \begin{bmatrix} I_2 & 0 \\ 0 & X \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

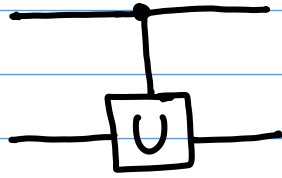
eg. $\begin{bmatrix} X & 0 \\ 0 & I_2 \end{bmatrix} = NOTC$

$$\begin{bmatrix} I_2 & 0 \\ 0 & Z \end{bmatrix} = CZ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

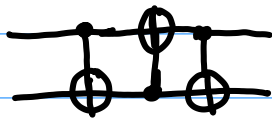
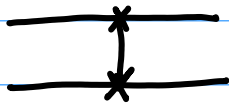
$$\begin{array}{c} \text{---} \\ | \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ | \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ | \\ \text{---} \end{array} = \begin{array}{c} \text{---} \\ | \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ | \\ \text{---} \end{array}$$

$$\begin{array}{cc|cc} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & - & 1 \end{array}$$

$$\text{Controlled-U} = \begin{bmatrix} I_2 & 0 \\ 0 & U_{2 \times 2} \end{bmatrix}$$

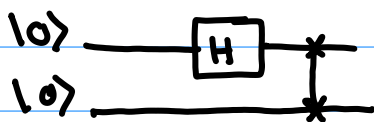


$$\text{SWap Gate} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$\begin{array}{cc|cc} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{array}$$

$$|01\rangle \Rightarrow |01\rangle \Rightarrow |11\rangle \Rightarrow |10\rangle$$



$$\begin{array}{l} |00\rangle \\ |1+0\rangle \\ |0+\rangle \end{array}$$

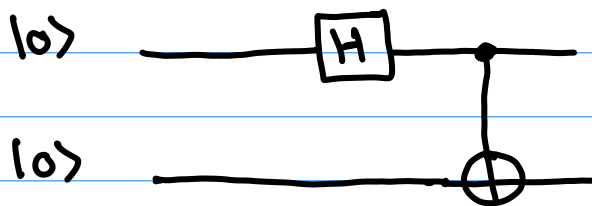
$$\begin{array}{l} H \text{ on } q_0 \\ \text{SWap} \end{array}$$

$$\frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |10\rangle \quad \text{SWAP}$$

$$= \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |01\rangle$$

$$= \frac{1}{\sqrt{2}} |0\rangle (|0\rangle + |1\rangle)$$

$$= |0\rangle |+\rangle$$



$$|00\rangle$$

H on q_0

$$\frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) \quad \text{CNOT } q_0, q_1$$

$$\frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

M q_0

$p(0) = 0.5 \Rightarrow q_1$ will be $|0\rangle$

$p(1) = 0.5 \Rightarrow q_1$ will be $|1\rangle$

Entanglement

Bell states:

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$$\frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$\frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

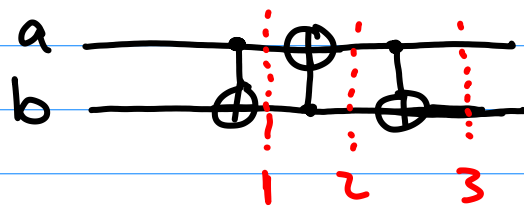
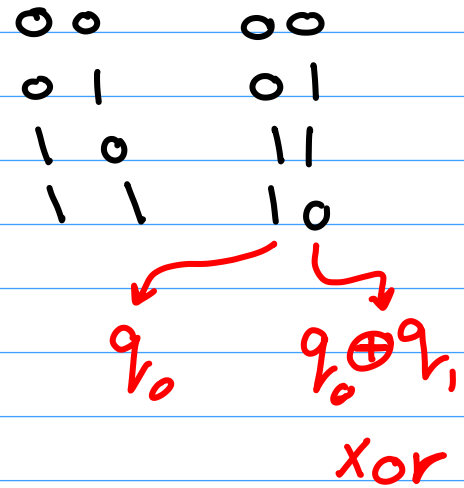
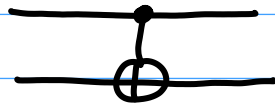
$$\frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$\frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

$$\begin{aligned} |\psi\rangle &= (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) \\ &= ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle \\ &= \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle \end{aligned}$$

$$\begin{array}{cccc} ac = \frac{1}{\sqrt{2}} & ad = 0 & bc = 0 & bd = \frac{1}{\sqrt{2}} \\ \swarrow \times & \downarrow & \downarrow & \nwarrow \times \\ & a = 0 & d = 0 & \end{array}$$

can't be presented with  
product states



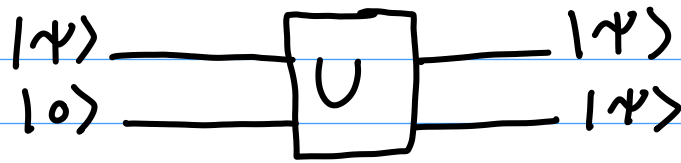
$$1: \begin{matrix} a \\ a \oplus b \end{matrix} \quad 2: \begin{matrix} a \oplus a \oplus b = b \\ a \oplus b \end{matrix}$$

$$3: \begin{matrix} b \\ a \oplus b \oplus a = a \end{matrix} \quad \text{Final: } \begin{matrix} b \\ a \end{matrix}$$

No clone theorem:

We cannot copy any quantum state  
only move it.

prove:



$$U |\psi\rangle |0\rangle = |\psi\rangle |\psi\rangle$$

this is  
irreversible



open control

